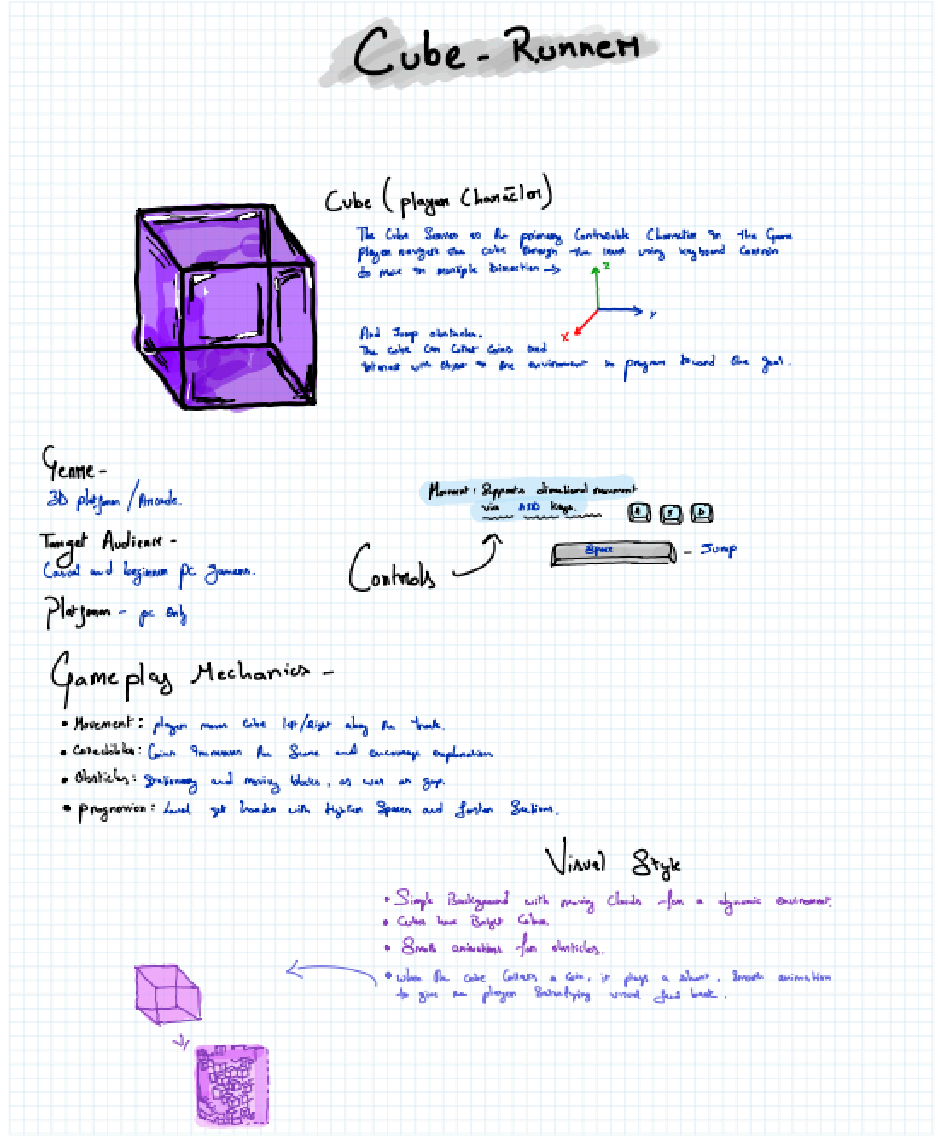
Concept Ideation (Think Without Limits)

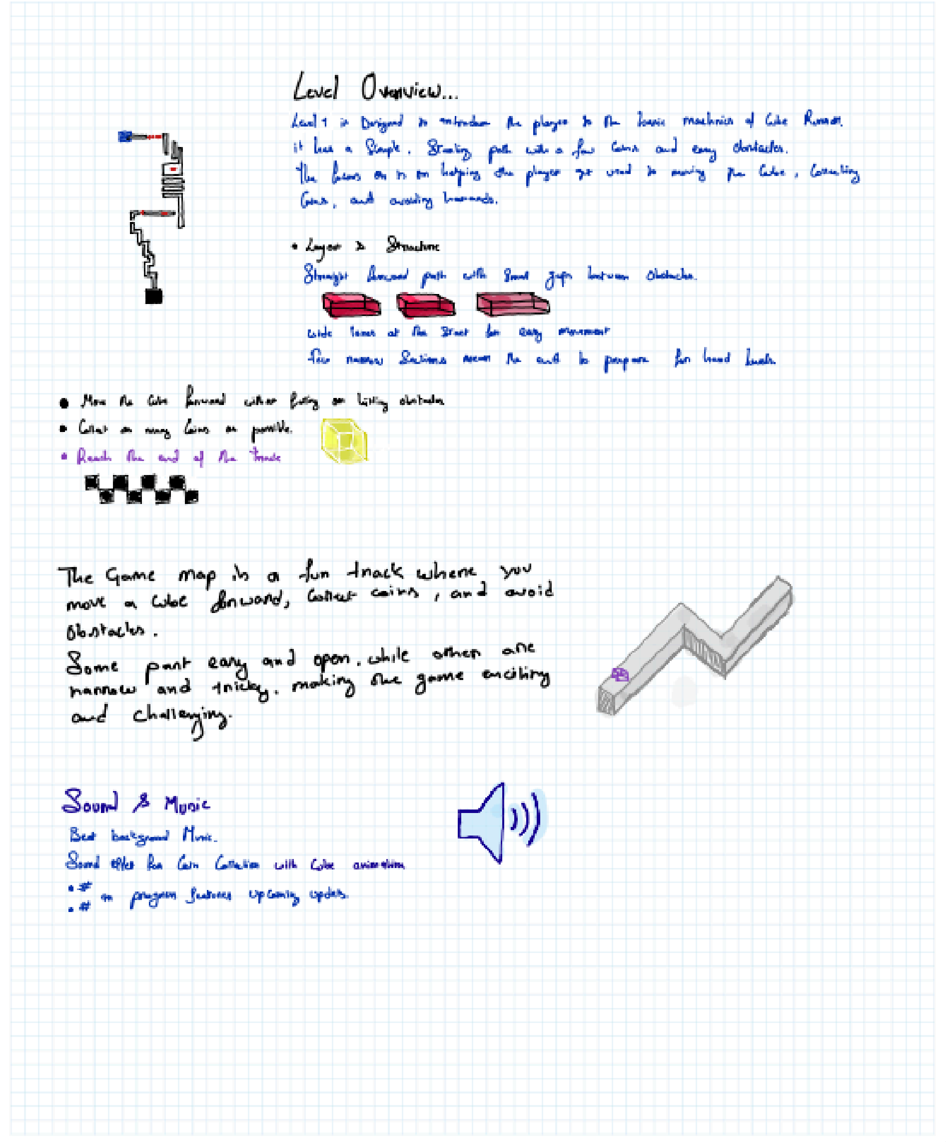
We began by exploring multiple ideas without restrictions. We aimed for a game that was visually simple but mechanically satisfying, leading to the initial concept of a geometric runner focused on movement, precision, and reflexes.

Scoping the Concept for Feasibility

To fit the one-week development timeline, we refined the idea into a level-based 3D runner with:

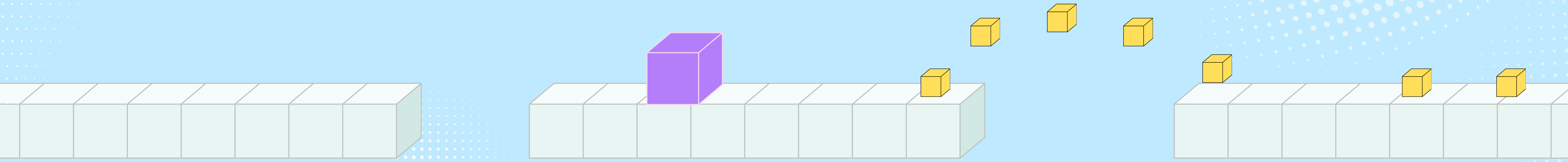
1. Simple controls (Move + Jump)
2. Three handcrafted levels
3. Three core obstacle types
4. Clean geometric visuals
5. This ensured the project was achievable while still offering variety.





CUBE RUNNER

Cube Runner is a 3D isometric game developed for the Windows platform. Unlike traditional endless runners, it features handcrafted levels, structured progression, and a thoughtfully balanced difficulty curve to deliver a more engaging and immersive gameplay experience.



This is an early prototype, and the gameplay and overall experience will continue to evolve. It should not be considered representative of the final product.

Gameplay Mechanics

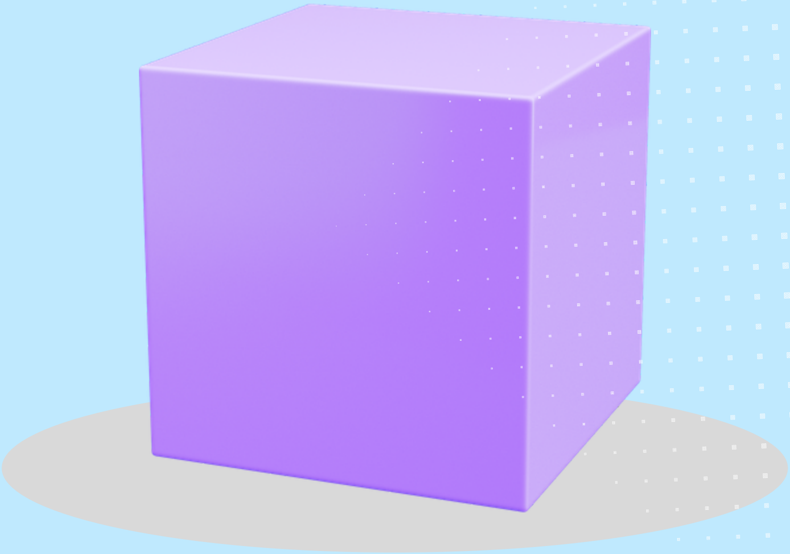
Player Character (The Controller)
Avatar: A vibrant Purple Cube.

Directional: The cube moves on the X and Z axes (Forward, Backward, Left, Right).

Physics: The cube utilizes rigid body physics (it rolls and tumbles slightly rather than sliding perfectly flat).

Jump: A vertical impulse force to clear gaps between platforms.

Controls
W / A / S / D: Move Directionally.
Space Bar: Jump.
R Key: Restart Level (Instant respawn).

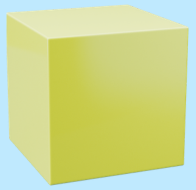


Objectives

Primary Goal: Navigate from the start of the path to the end without falling off the map.

Secondary Goal (Scoring): Collect Yellow Cube Coins scattered along the path.

Score System: +1 Point for every coin collected. The score is displayed in real-time on the screen.



Coin

Failure State

If the player falls off the platform (Y-axis drops below a certain threshold).
The player press 'R' to restart the level from the beginning.

Obstacles & Hazards

Static Red Cube

- Touch = restart.
- Introduced: Level 1.

Moving Hazards

- Move in predefined patterns (L/R/U/D).
- Touch = restart.
- Introduced: Level 2.

Falling Platforms

- Drop when player approaches.
- Introduced: Level 3.

UPGRADABLE OBSTICLES

Disappearing Tiles

Tiles blink (visible → invisible). Player must move when the tile is solid.

Speed Boost Pads

Blue pads that push the player forward very fast, increasing difficulty in jump timing.

Wind Push Zones

Invisible force pushes the cube slightly left/right, adding instability.

Rotating Bar Obstacle

A long bar rotates on the platform, forcing the player to time a jump or move to the safe side.

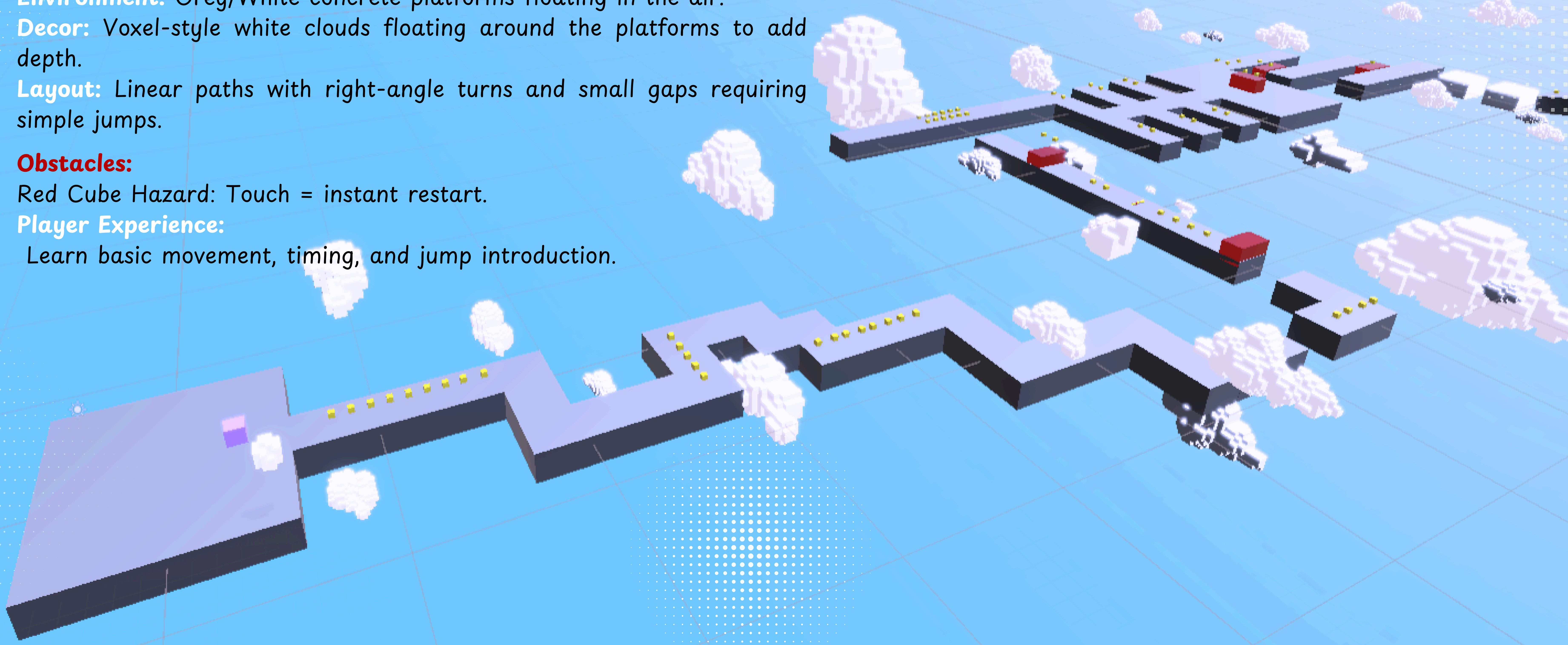
Disappearing Tiles

Tiles blink (visible → invisible). Player must move when the tile is solid.

Level Design & Biomes

Level 1: The Sky Realm
Visuals: Bright blue sky background.
Environment: Grey/White concrete platforms floating in the air.
Decor: Voxel-style white clouds floating around the platforms to add depth.
Layout: Linear paths with right-angle turns and small gaps requiring simple jumps.

Obstacles:
Red Cube Hazard: Touch = instant restart.
Player Experience:
Learn basic movement, timing, and jump introduction.

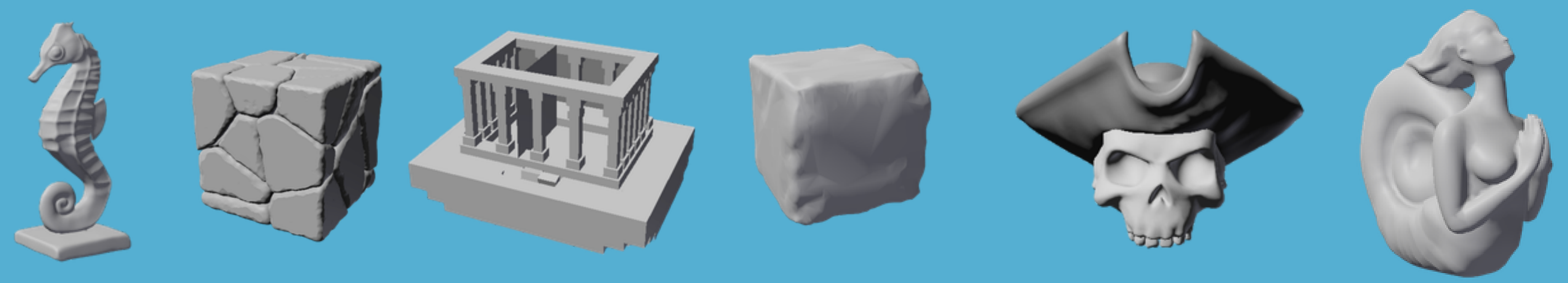


Level Design & Biomes

Level 2: The Water Ruins
Visuals: Bright turquoise water shader with reflections/caustics.
Environment: Ancient grey stone tiles/bricks.
Decor: Ruined stone pillars, giant stone heads, and submerged blocks.
Layout: A maze-like structure sitting just above the water surface.

Obstacles:
Moving Platforms/Hazards: Move left/right/up/down.
Touching them restarts the level
Player Experience:
Improve timing precision with dynamic movement patterns.

3D models identified in the prototype that constitute the level's geometry.

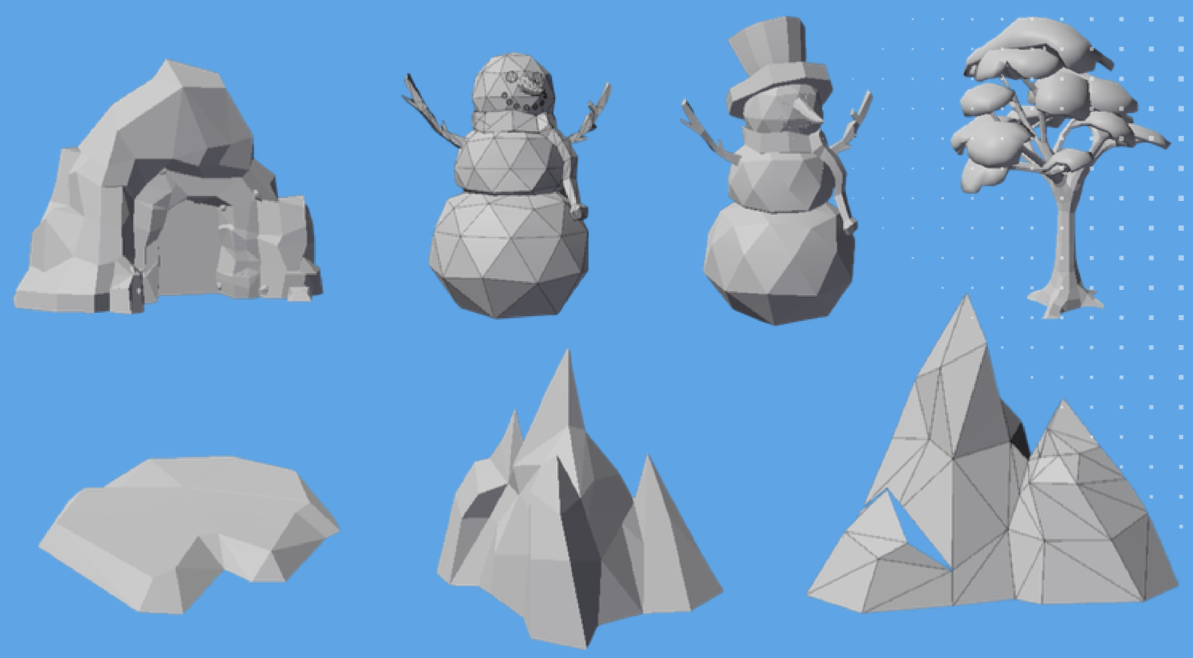


Level Design & Biomes

Level 3: The Ice World (Video 3)
Visuals: Cold, foggy white/grey atmosphere.
Environment: Light blue "Ice" platforms snow everywhere..
Decor: Low-poly pine trees and snowmen (with top hats and scarves) placed on the platforms.
Layout: winding paths with larger gaps.

Obstacles:
Falling Platforms: Trigger when the player gets close.
 Ground collapses within 1-2 seconds.
Player Experience:
Master speed, timing, and fast decision-making.

3D models identified in the prototype that constitute the level's geometry.



Map Path Architecture

General Layout The map paths follow a "Linear Gauntlet" design. There is a single, defined route from start to finish with no branching paths. This forces the player to master specific obstacles in a set order. The path is constructed using a grid-based tile system, ensuring predictable movement metrics.

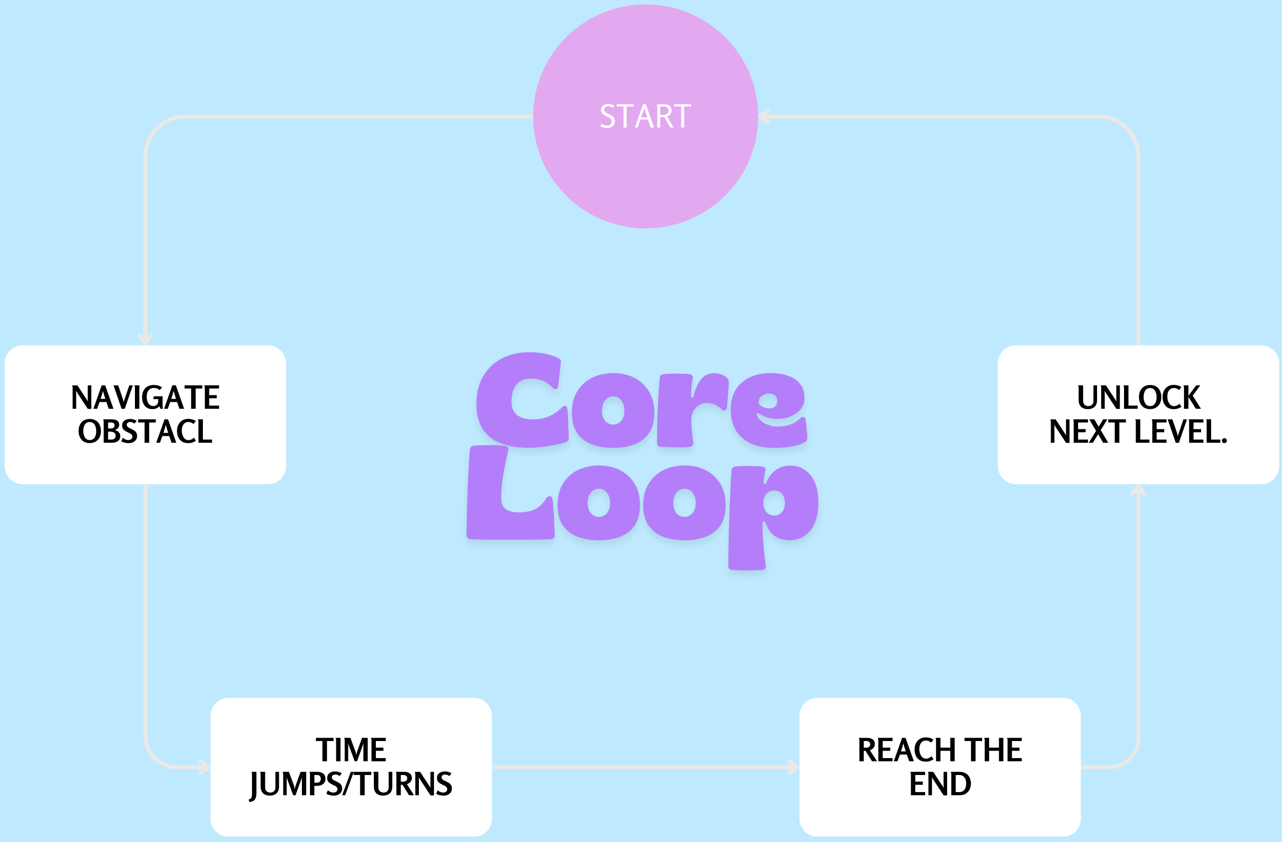
Avatar System

Feature: Players can swap the default "Cube" for one of 5 unique 3D characters.

Roster:

- Default: Standard futuristic bot.
- Desert Theme: Matches the Sandstone level.
- Ice Theme: Matches the Glacial level.
- Aerodynamic: Sleek design for speed.
- Gold/Premium: A reward for high scores.

System: Characters are unlocked using Coins collected in levels and equipped via a "Shop" menu. The hitbox remains the same for all avatars to keep gameplay fair.



Cube Runner is a 3D isometric platformer developed for Windows, featuring handcrafted levels instead of endless-generation. The game focuses on tight movement, precise timing, and learning-based progression. Players navigate three unique biomes—Ocean, Antarctic, and Sky Isles—each introducing new obstacles, mechanics, and visual identities. Core mechanics include turning, jumping, and avoiding hazards that instantly restart the level on contact.

Design began with broad concept exploration, then was scoped down and structured using a beat chart to define level flow, difficulty pacing, new mechanics, and player objectives. Each level teaches timing, coordination, and environmental awareness through escalating challenges such as static danger blocks, moving platforms, and falling-path traps. The goal of every stage is simple: survive from start to finish while mastering the rhythm and patterns of each environment.